

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Engineering (UE24CS341A) — Mini Project (Jackfruit Phase-1)

PROJECT SYNOPSIS

1. PROJECT TITLE AND METADATA

Field	Details
Project Title	PassPulse – Smart Event Discovery, High-Concurrency Ticketing & Anti-Fraud Check-in Platform
Group Number	Team 2
Project ID	PP-2026-T02
Customer / Stakeholders	Independent Event Organizers (Conferences, Cultural Shows, Tech Meetups), Campus Clubs, Venue Managers & Attendees
Technology Stack	React.js (SPA), Node.js (Express.js), PostgreSQL (ACID), Redis (Atomic Locks & Caching)
Complexity Level	Medium (Advanced Concurrency, Cryptographic Validation & Real-Time Sync)

2. PROBLEM STATEMENT

Modern event management across commercial concerts, professional conferences, exhibitions, and university festivals suffers from fragmented tooling and critical operational vulnerabilities:

- Ticket Duplication & Scalping:** Static PDF tickets and screenshot-shared QR codes lead to duplicate entries and fraudulent gate admissions.
- Race Conditions During Peak Bookings:** Traditional booking systems crash or cause inventory overselling when hundreds of users attempt to purchase limited tickets simultaneously.
- Uncontrolled Capacity & Empty Seats:** When events sell out, potential attendees are turned away without an automated waitlist mechanism, leaving venues under-capacity when cancellations occur.
- Gate Congestion & Network Latency:** Manual list cross-referencing and slow server round-trips cause massive ingress queues and entry delays at venue gates.
- Lack of Real-Time Operational Visibility:** Organizers lack live telemetry regarding entry throughput, attendance drop-off, and ticket tier performance.

PassPulse solves these challenges by combining dynamic cryptographic QR verification, ACID-compliant ticket locking, automated FIFO waitlist reallocation, and live telemetry in an integrated, production-grade web platform.

3. PROJECT OBJECTIVES

- High-Concurrency Ticket Reservation:** Implement atomic inventory locking with a 10-minute Time-To-Live (TTL) reservation hold to completely eliminate race conditions and overselling.
- Anti-Fraud Cryptographic Check-in:** Issue dynamic, time-synced, HMAC-signed QR tickets that prevent screenshot forwarding and duplicate scans.
- Zero-Waste Automated Waitlisting:** Auto-queue users for sold-out tiers and implement an automated 15-minute claim-window reallocation engine upon ticket cancellations.
- Sub-200ms Gate Verification:** Provide an optimized, camera-based gate scanning interface capable of sub-200ms cryptographic validation and duplicate rejection.
- Live Attendance Telemetry & Analytics:** Deliver real-time dashboards tracking check-in velocity (scans/min), live occupancy heatmaps, and exportable financial/attendance audit reports.
- Full SDLC & CI/CD Compliance:** Deliver the project using strict Agile sprints, GitFlow, containerized microservices (Docker), automated testing ($\geq 80\%$ coverage), and SonarQube quality gates.

4. USER CLASSES AND PERSONA ROLES

- **Attendee / General Consumer:** Discovers and filters events; reserves tickets across multi-tiered quotas (Early Bird, General, VIP); manages dynamic QR passes; joins automated waitlists for sold-out events.
- **Event Organizer / Host:** Designs multi-tier ticket categories; sets quotas, schedules, and custom forms; verifies attendee QR passes via in-app scanner; monitors real-time check-in velocity and post-event analytics.
- **Gate Staff / Check-in Agent:** Dedicated scanning role with high-speed camera validation access, visual success/error cues, and manual override capabilities for registered exceptions.
- **Platform Super-Admin:** Reviews and approves event proposals; monitors platform-wide financial/traffic audits; handles user governance and enforces trust/safety compliance.

5. ITEMIZED FUNCTIONAL FEATURES

- 1. **User Authentication, Session & Role-Based Access Control (RBAC):** Multi-tier registration and login with bcrypt password hashing, JWT stateless authentication, refresh token rotation, and strict role permissions (Attendee, Organizer, Gate Staff, Admin).
- 2. **Smart Event Discovery & Multi-Faceted Catalog:** Elastic-style event discovery supporting compound filters (category, date range, location, price range, availability) with responsive card layouts and rich event detail views.
- 3. **Event Lifecycle & Workflow Management:** Comprehensive dashboard enabling organizers to draft, publish, schedule, modify, and cancel events, with multi-stage approval states (Draft, Under Review, Published, Live, Completed).
- 4. **Multi-Tier Ticket Inventory & Pricing Engine:** Granular configuration of ticket tiers (Early Bird, VIP, General Admission, Student) with independent pricing, capacity caps, release schedules, and per-user purchase limits.
- 5. **Atomic Ticket Booking with TTL Hold Engine:** Concurrency-safe ticket checkout utilizing Redis-backed temporary locks (10-minute checkout window) and PostgreSQL ACID transactions to guarantee zero overbooking.
- 6. **Dynamic Anti-Fraud QR Pass Issuance:** Generation of cryptographic, HMAC-signed, time-expiring digital QR passes accessible via attendee dashboard, preventing static screenshot sharing.
- 7. **Fast Gate Check-in & Duplicate Detection:** Mobile camera-enabled scanner module that verifies cryptographic QR signatures, instantly flags used/invalid passes, and records gate entry timestamps in under 200ms.
- 8. **Automated Waitlist & FIFO Slot Reallocation:** Automated queuing system for sold-out events that automatically notifies and offers reserved slots to waitlisted users with a 15-minute claim countdown timer.
- 9. **Real-Time Attendance Telemetry & Post-Event Analytics:** Live dashboard displaying entry throughput graphs, attendance vs. registration percentages, revenue breakdown per tier, and downloadable CSV/PDF audit reports.
- 10. **Platform Administration & Compliance Moderation:** Super-admin portal for reviewing pending event submissions, moderating inappropriate listings, resolving disputed transactions, and viewing platform-wide KPIs.

6. PRODUCT OWNERSHIP & TEAM RESPONSIBILITY MATRIX

Note: In compliance with SE guidelines, each member owns one distinct, end-to-end functional feature across all SDLC phases. UI design, DB modeling, and CI/CD pipelines are allocated as secondary engineering responsibilities.

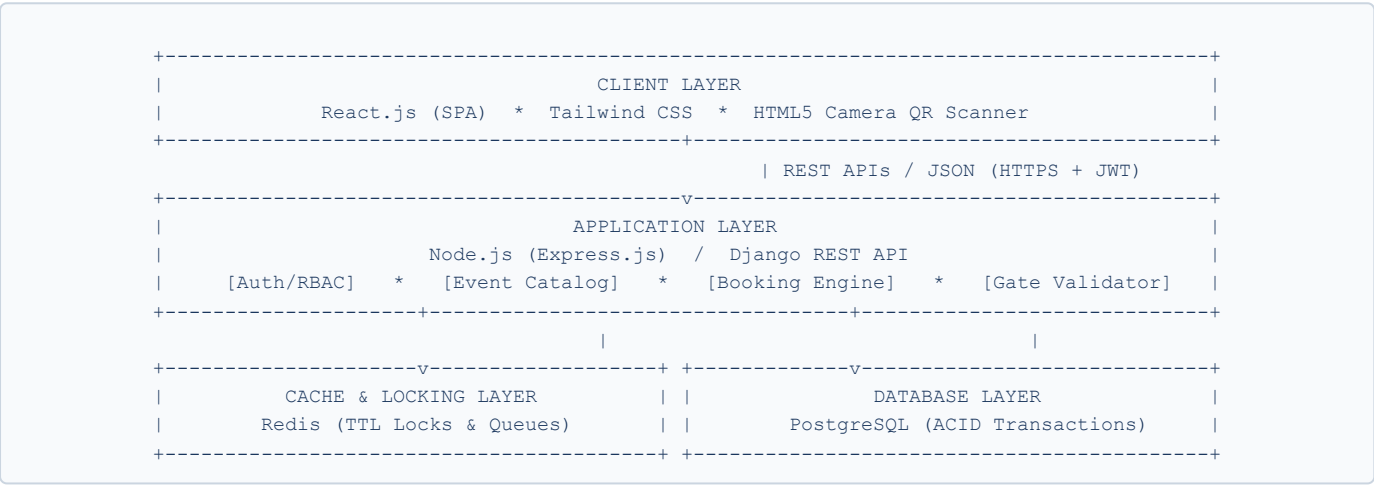
Member	Primary Owned Functional Feature	Secondary Engineering Responsibility	Qualitative Engineering Goal (SLA)
Member 1 (Lead)	Authentication, RBAC & Platform Governance Module	API Gateway, Middleware & Security Hardening	Zero High/Medium vulnerabilities in SonarQube; sub-80ms authentication token validation.
Member 2	Event Catalog, Multi-Faceted Discovery & Lifecycle Engine	PostgreSQL Database Schema & Compound Indexing	Sub-120ms search response time across 10,000+ mock indexed records using composite indexes.
Member 3	Atomic Ticket Booking, TTL Inventory Lock & Waitlist Reallocation	Redis Concurrency, Distributed Locking & Transactions	100% data consistency with 0 overbooking under 200 concurrent checkout requests.
Member 4	Cryptographic QR Pass Generation, Gate Scanner & Telemetry Engine	Testing Framework (Jest/Supertest) & CI/CD Pipeline	Sub-200ms QR scan verification latency; ≥ 85% branch test coverage in automated CI builds.

7. SHORT-TERM AGILE PLAN (SPRINT ROADMAP)

Iteration	Duration	Planned Engineering Output	Acceptance & Verification Evidence
Sprint 1	Weeks 1–2	SRS baseline, Jira user stories mapping, Git repository setup with branch protections, Docker environment.	Approved SRS v1.0 + Configured GitHub repo + Working Docker-Compose scaffolding.

Sprint 2	Weeks 2–3	Authentication API, RBAC middleware, User Profiles, and Database migrations.	Working Auth endpoints + Unit tests + Passing SonarQube quality gate.
Sprint 3	Weeks 3–4	Event creation wizard, multi-tier pricing setup, and catalog search engine.	Catalog UI with filter queries + Swagger API documentation + Integration tests.
Sprint 4	Weeks 5–6	Atomic ticket reservation with TTL lock, mock checkout, and automated waitlist engine.	Automated concurrency test scripts proving zero inventory overselling.
Sprint 5	Weeks 6–7	Dynamic QR generation, camera gate scanner module, and real-time attendance telemetry.	Live camera QR verification demo + Gate log verification + ≥ 80% test coverage.
Sprint 6	Weeks 7–8	Admin moderation portal, report generation, end-to-end integration testing, and CI/CD deployment.	Automated Jenkins CI pipeline execution + Final live demo of complete event lifecycle.

8. SYSTEM ARCHITECTURE & INFRASTRUCTURE PIPELINE



- **Core Stack:** React.js, Node.js (Express.js), PostgreSQL, Redis, Tailwind CSS.
- **DevOps & CI/CD:** Git, GitHub (PR reviews), Jenkins / GitHub Actions, Docker & Docker-Compose.
- **Quality Assurance:** SonarQube (Static security & code smell analysis), Jest & Supertest (≥ 80% coverage), Postman.
- **Project Management:** Jira (Agile Epics, User Stories, Sprint Backlogs, Velocity charts).

9. SCOPE & BOUNDARY CONDITIONS

In Scope: Complete end-to-end event lifecycle (creation, approval, publishing); multi-tier ticket limits; atomic checkout with 10-min reservation hold; dynamic cryptographic QR passes; mobile-friendly camera gate verification; automated 15-min claim waitlist reallocation; live attendance dashboards and CSV export.

Out of Scope (Explicit Non-Goals): Real monetary payment gateway settlement (mock payment tokens will be used); facial recognition entry; live video streaming; GPS venue indoor maps.